At emission, the satellite distances to both stations are equal:

When satellite is in the triangle plane, its altitude is maximal, . Let be the latitude of . The cosine rule gives

Equations give

Equation also follows directly from the diagram.

A small latitude error gives, by , an error in . Then we have

Expand . Small gives small . Drop higher orders and subtract , giving

Here is given by .

An error in time gives an error in :

By , this gives an error in . Then

From and , neglect higher orders to obtain

Here is given by .

From , From ,

Substituting into gives

. In a chosen frame, let position be and time . Satellite has coordinates at time . Its signal travels at light speed, so

has four unknowns , requiring four independent equations. Receive four or more satellites at once. Position and time follow from ,